

05 The Network Layer

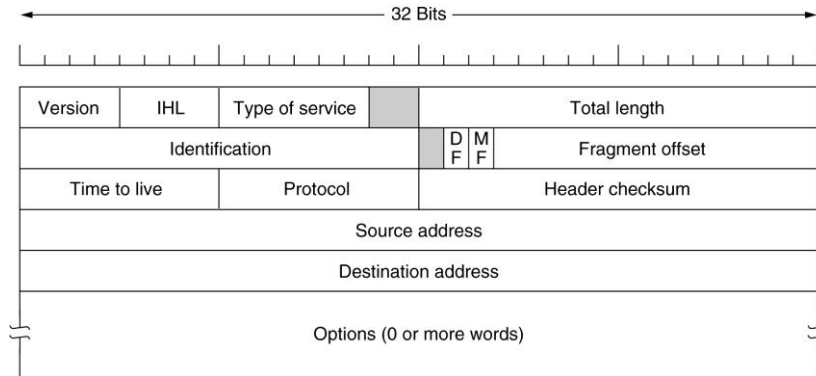
05.01 The Network layer in the Internet – IP Protocol

The Network Layer in the Internet

- The IP Protocol
- IP Addresses
- Internet Control Protocols
- IPv6

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The IP Protocol



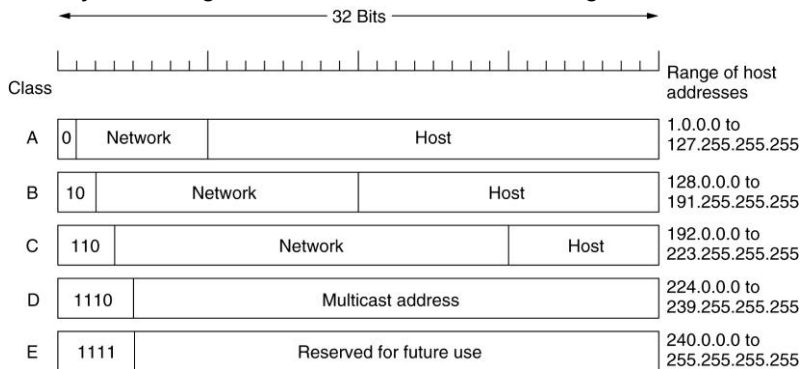
The IP Protocol (2)

Some of the IP options.

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

IP Addresses

- This is the original IP address format
- It is based on *class addressing*
- Today is no longer used → see CIDR addressing later



IP address formats.

IP Addresses

- The main problem with this type of addressing is the unbalancing of nr. of hosts with respect to the nr. of possible sub-networks (e.g. Ethernet)
- Ex. B class addresses: 64000 hosts may be too much for a single organization (company, faculty, etc.)

0 0	This host
0 0 . . . 0 0 Host	A host on this network
1 1	Broadcast on the local network
Network 1 1 1 1 . . . 1 1 1 1	Broadcast on a distant network
127 (Anything)	Loopback

Special IP addresses.

Special Addresses

- 0.0.0.0 current host without specifying address;
- 255.255.255.255 limited broadcast address;
- 127.x.y.z loopback address: datagrams will be delivered to upward layers of the same host.
- 169.254.x.y host autoconfig.

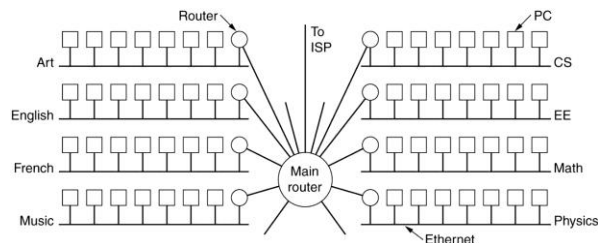
Private IP

- Three blocks, one for each class.
- Class A: 10.0.0.0
- Class B: 172.16.0.0 - 16 contiguous networks;
- Class C: 192.168.0.0 - 255 contiguous networks.

Classe A	10.0.0.0	a 10.255.255.255
Classe B	172.16.0.0	a 172.31.255.255
Classe C	192.168.0.0	a 10.255.255.255

Subnets

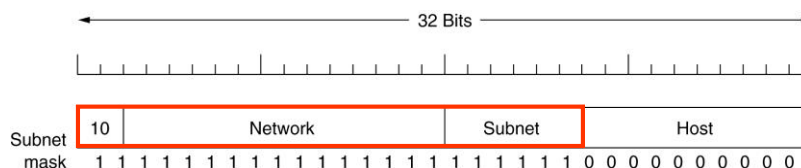
- Instead of 1 network of 65536 hosts it may be useful to have 64 Ethernet with 1022 hosts
- This can be achieved by means of a **subnet mask** in order to assign, for instance, 6 of the 16 bits devoted to hosts to the subnet and the remaining to the hosts



A campus network consisting of LANs for various departments.

Subnets

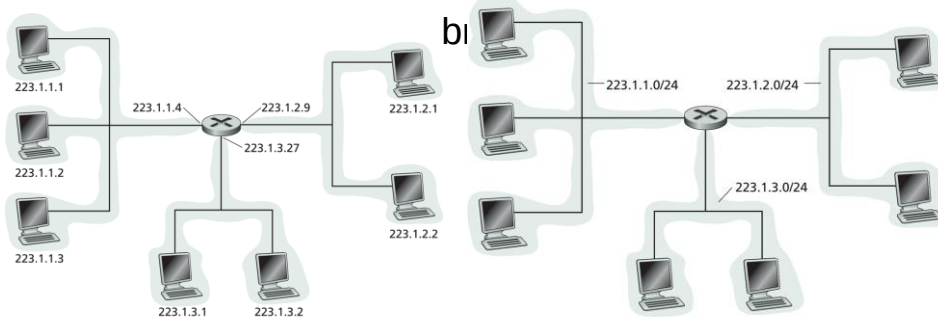
- The subnet mask marks the boundary between network and host addresses
- In the example below: 255.255.252.0 (also denoted by /22 in CIDR) is the subnet mask



A class B network subnetted into 64 subnets.

Subnets

- Each interface of host/router on the Internet has a unique IP address
- 223.1.1.1,.2,.3,.4 are a subnet



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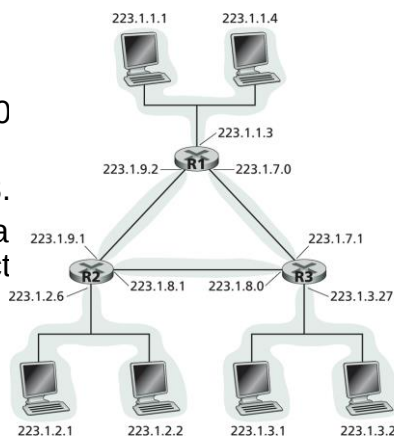
Interface addresses and subnets

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♦ Subnet addresses

Subnets

- Each router has 3 interfaces
- We have 3 subnets: 223.1.1.0
223.1.3.0/24
- But also 2 more subnets: 223.
- A subnet can hence defined a
whose end points are connect
interface



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♦ Three routers interconnecting six subnets

CIDR – Classless InterDomain Routing

- IP addresses are assigned based on the generalization of subnetwork addressing
- IP addresses are divided in 2 parts: **a.b.c.d/x**
- x specifies the nr. of bits of the **prefix**
- Routing is hierarchical since:
 - Router external to the network of the organization need only to store a record of type a.b.c.d/x
 - Internal routers are in charge of using remaining bit to deliver packets
 - Remaining bit can in turn be used to denote subnets internal to the organization

IPv6 Adoption

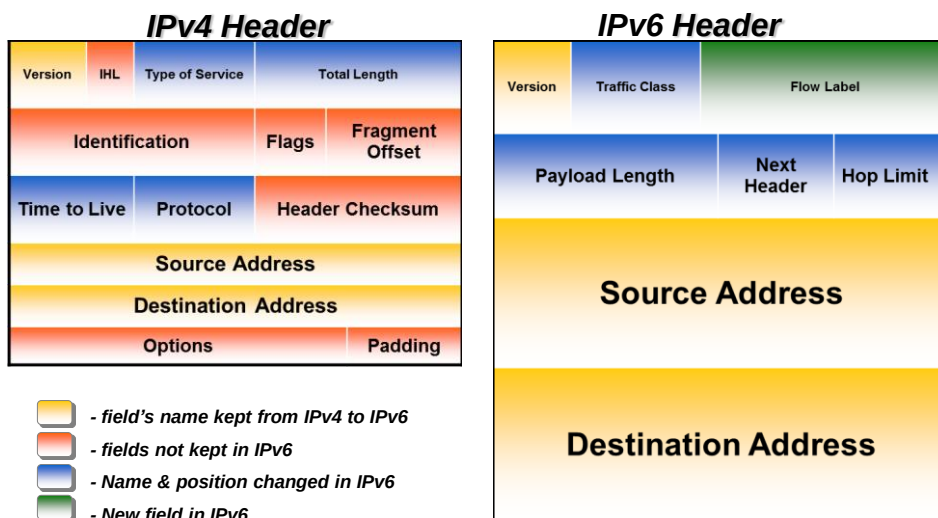
	Country/Region	Q3 2016 IPv6 %	▼ RANK	IPv6%	COUNTRY
			1	66%	India
1	Belgium	39%	2	63.2%	Saint Barthelemy
2	Greece	25%	3	55.4%	Mayotte
3	Germany	22%	4	45%	United States
4	United States	21%	5	43.8%	Belgium
5	Switzerland	21%	6	42.5%	Malaysia
6	Ecuador	20%	7	42.2%	Viet Nam
7	India	17%	8	40.7%	Germany
8	Portugal	16%	9	40.2%	Greece
9	Estonia	15%	10	37.5%	French Guiana
10	Peru	12%			

Figure 4: IPv6 Traffic Percentage, Top Countries/Reg.....

IPv6 - So what's really changed ?!

- Expanded Address Space
 - Address length quadrupled to 16 bytes
- Header Format Simplification
 - Fixed length, optional headers are daisy-chained
 - IPv6 header is twice as long (40 bytes) as IPv4 header without options (20 bytes)
- No checksumming at the IP network layer
- No hop-by-hop segmentation
 - [Path MTU discovery](#)
- 64 bits aligned
- Authentication and Privacy Capabilities
 - IPsec is mandated
- No more broadcast

IPv4 & IPv6 Header Comparison



Extension Headers

- Since some of the eliminated IPv4 fields can still be required, the concept of extension headers has been introduced

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

IPv6 extension headers.

How Was IPv6 Address Size Chosen?

- Some wanted fixed-length, 64-bit addresses
 - Easily good for 10^{12} sites, 10^{15} nodes, at .0001 allocation efficiency (3 orders of magnitude more than IPv6 requirement)
 - Minimizes growth of per-packet header overhead
 - Efficient for software processing
- Some wanted variable-length, up to 160 bits
 - Compatible with OSI NSAP addressing plans
 - Big enough for auto-configuration using IEEE 802 addresses
 - Could start with addresses shorter than 64 bits & grow later
- Settled on fixed-length, 128-bit addresses
 - (340,282,366,920,938,463,463,374,607,431,768,211,456 in all!)

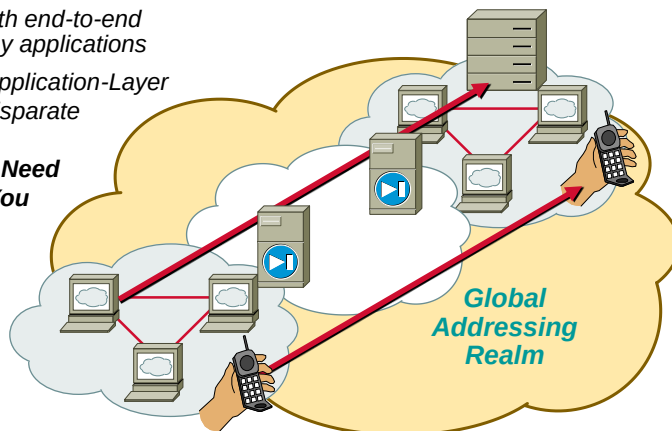
Benefits of 128 bit Addresses

- Room for many levels of structured hierarchy and routing aggregation
- Easier address management and delegation than IPv4
- Easy address auto-configuration
- Ability to deploy end-to-end IPsec (NATs removed as unnecessary)

Coming Back to an End-to-End Architecture

New Technologies/Applications for Home Users 'Always-on'—Cable, DSL, Ethernet-to-the-home, Wireless,...

- Internet started with end-to-end connectivity for any applications
 - Today, NAT and Application-Layer Gateways connect disparate networks
- Always-on Devices Need an Address When You Call Them, eg.**
- Mobile Phones
 - Gaming
 - Residential Voice over IP gateway
 - IP Fax



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IPv6 Technology Scope			
IP Service	IPv4 Solution	IPv6 Solution	
Addressing Range	32-bit, Network Address Translation	128-bit, Multiple Scopes	
Autoconfiguration	DHCP	Serverless, Reconfiguration, DHCP	
Security	IPSec	IPSec Mandated, works End-to-End	
Mobility	Mobile IP	Mobile IP with Direct Routing	
Quality-of-Service	Differentiated Service, Integrated Service	Differentiated Service, Integrated Service	
IP Multicast	IGMP/PIM/Multicast BGP	MLD/PIM/Multicast BGP, Scope Identifier	
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IPv6 Addressing			
- IPv6 Addressing rules are covered by multiples RFC's - Architecture defined by RFC 3513 (obsoletes RFC 2373) - Address Types are : - Unicast : One to One (Global, Link local, Site local, Compatible) - Anycast : One to Nearest (Allocated from Unicast) - Multicast : One to Many - Reserved - A single interface may be assigned multiple IPv6 addresses of any type (unicast, anycast, multicast) - No Broadcast Address -> Use Multicast			
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IPv6 Address Representation

- 16-bit fields in case insensitive colon hexadecimal representation
 - 2031:0000:130F:0000:0000:09C0:876A:130B
- Leading zeros in a field are optional:
 - 2031:0:130F:0:0:9C0:876A:130B
- Successive fields of 0 represented as ::, but only once in an address:
 - 2031:0:130F::9C0:876A:130B
 - 2031::130F::9C0:876A:130B
 - 0:0:0:0:0:0:1 => ::1
 - 0:0:0:0:0:0:0 => ::
- IPv4-compatible address representation
 - 0:0:0:0:0:0:192.168.30.1 = ::192.168.30.1 = ::C0A8:1E01

Unicast Address Format

Link Local

FP (10bits)	RESERVED (54bits)	Interface ID (64bits)
1111111010	MUST be 0	MAC derived

Site Local

FP (10bits)	Subnet (38bits)	Subnet (16bits)	Interface ID (64bits)
1111111011	Locally Administered	Locally Administered	MAC derived or Locally Administered

Global

FP (3bits)	Registry/ provider assigned (45bits)	Subnet (16bits)	Interface ID (64bits)
001	Provider Administered	Locally Administered	MAC derived or Locally Administered or Random

Some Special-Purpose Unicast Addresses

- The unspecified address, used as a placeholder when no address is available:

0:0:0:0:0:0:0:0

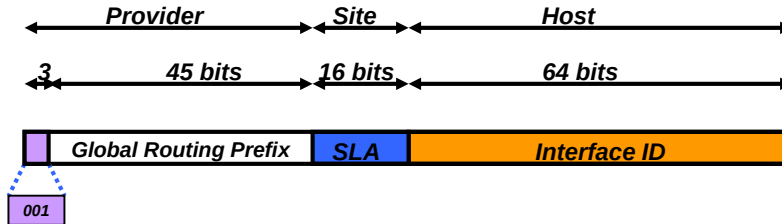
- The loopback address, for sending packets to self:

0:0:0:0:0:0:0:1

IPv6 Addressing

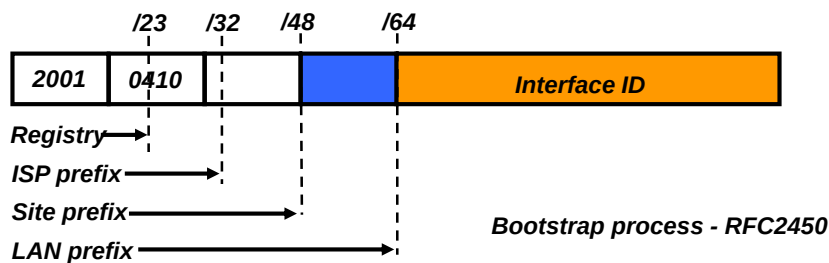
- Prefix Format (PF) Allocation
 - PF = 0000 0000 : Reserved
 - PF = 001 : Aggregatable Global Unicast Address
 - PF = 1111 1110 10 : Link Local Use Addresses (FE80::/10)
 - PF = 1111 1110 11 : Site Local Use Addresses (FEC)::/10)
 - PF = 1111 1111 : Multicast Addresses (FF00::/8)
 - Other values are currently Unassigned (approx. 7/8th of total)
- All Prefix Formats have to support EUI-64 bits Interface ID setting
 - But Multicast

Aggregatable Global Unicast Addresses



- Aggregatable Global Unicast addresses are:
 - Addresses for generic use of IPv6
 - Structured as a hierarchy to keep the aggregation
- See RFC 3513

Address Allocation Policy



- The allocation process is under reviewed by the Registries:
 - IANA allocates 2001::/16 to registries
 - Each registry gets a /23 prefix from IANA
 - Formely, all ISP were getting a /35
 - With the new policy, Registry allocates a /32 prefix to an IPv6 ISP
 - Then the ISP allocates a /48 prefix to each customer (or potentially /64)
 - <ftp://ftp.cs.duke.edu/pub/narten/ietf/global-ipv6-assign-2002-06-26.txt>

Interface ID

- Lowest-order 64-bit field of unicast address may be assigned in several different ways:
 - auto-configured from a 64-bit EUI-64, or expanded from a 48-bit MAC address (e.g., Ethernet address)
 - auto-generated pseudo-random number (to address privacy concerns)
 - assigned via DHCP
 - manually configured

IPv6 Address Privacy (RFC 3041)



- Temporary addresses for IPv6 host client application, eg. Web browser
 - Inhibit device/user tracking but is also a potential issue
 - More difficult to scan all IP addresses on a subnet but port scan is identical when an address is known
 - Random 64 bit interface ID, run DAD before using it
 - Rate of change based on local policy
 - Implemented on Microsoft Windows XP
 - From RFC 3041: "...interface identifier ...facilitates the tracking of individual devices (and thus potentially users)..."

IPv6 Security

- All implementations required to support authentication and encryption headers (“IPsec”)
- Authentication separate from encryption for use in situations where encryption is prohibited or prohibitively expensive
- Key distribution protocols are under development (independent of IP v4/v6)
- Support for manual key configuration required

IP Quality of Service Approaches

Two basic approaches developed by IETF:

- “Differentiated Service” (diff-serv)
 - coarse-grain (per-class), qualitative promises (e.g., higher priority), no explicit signaling
- “Integrated Service” (int-serv)
 - fine-grain (per-flow), quantitative promises (e.g., x bits per second), uses RSVP signaling

IPv6 Support for Diff-Serv

8-bit Traffic Class field to identify specific classes of packets needing special QoS

- same as dscp definition of IPv4 Type-of-Service byte
- may be initialized by source or by router enroute; may be rewritten by routers enroute
- traffic Class value of 0 used when no special QoS requested (the common case today)

IPv6 Support for Int-Serv

20-bit Flow Label field to identify specific flows needing special QoS

- each source chooses its own Flow Label values; routers use Source Addr + Flow Label to identify distinct flows
- Flow Label value of 0 used when no special QoS requested (the common case today)
- this part of IPv6 is not standardized yet, and may well change semantics in the future

Compromise

- Signaled diff-serv (RFC 2998)
 - uses RSVP for signaling with course-grained qualitative aggregate markings
 - allows for policy control without requiring per-router state overhead